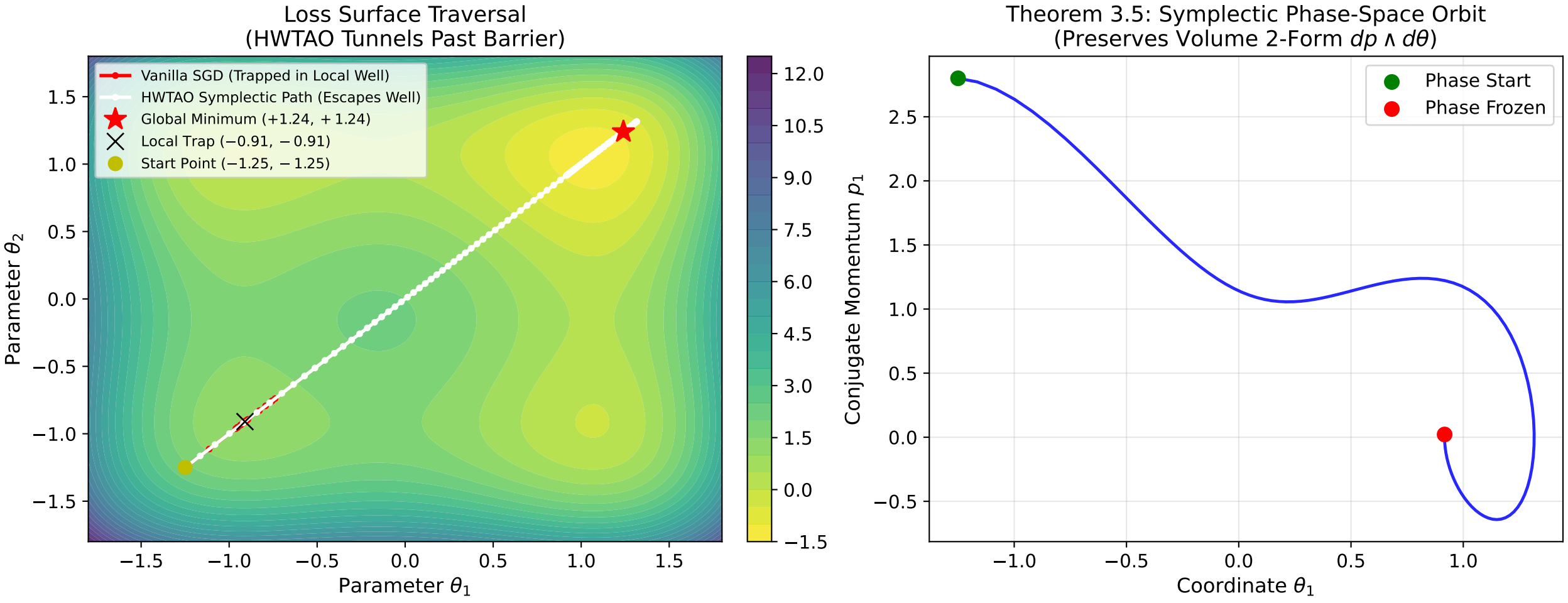

Loss Surface Traversal
(HWTAO Tunnels Past Barrier)
Vanilla SGD (Trapped in Local Well)
HWTAO Symplectic Path (Escapes Well)
Global Minimum (+1.24, + 1.24)
Local Trap (−0.91, − 0.91)
Start Point (−1.25, − 1.25)
Parameter θ_2
Parameter θ_1
Theorem 3.5: Symplectic Phase-Space Orbit
(Preserves Volume 2-Form $dp \wedge d\theta$)
Phase Start
Phase Frozen
Conjugate Momentum p_1
Coordinate θ_1